YTU
CIVIL ENGINEERING DEPARTMENT
TRANSPORTATION DIVISION

INS3441 HIGHWAY ENGINEERING LECTURE NOTES

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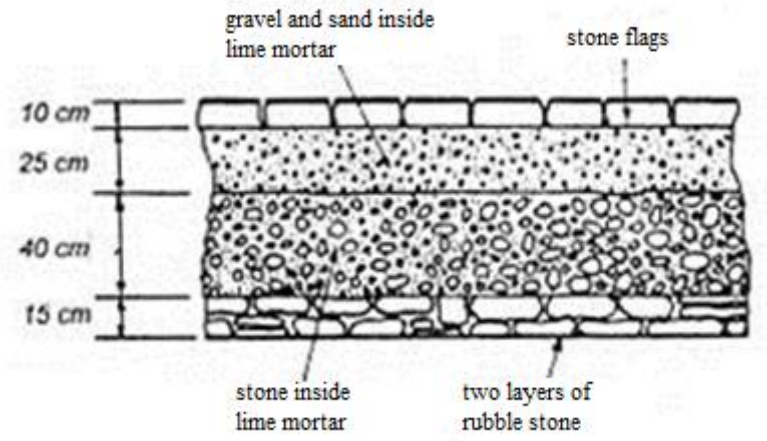
HISTORY OF HIGHWAY ENGINEERING

The history of road construction is parallel to the start of history of humanity. As people had to move because of their various needs, this result arised.

In the beginning, people formed pathways for their movement and then used sleds to carry their goods. Sleds were invented to carry heavy goods by spending less power. Then, tracks were formed to move sleds on smoother surfaces. As a result, the surface of the earth started to be modified to the desired objectives.

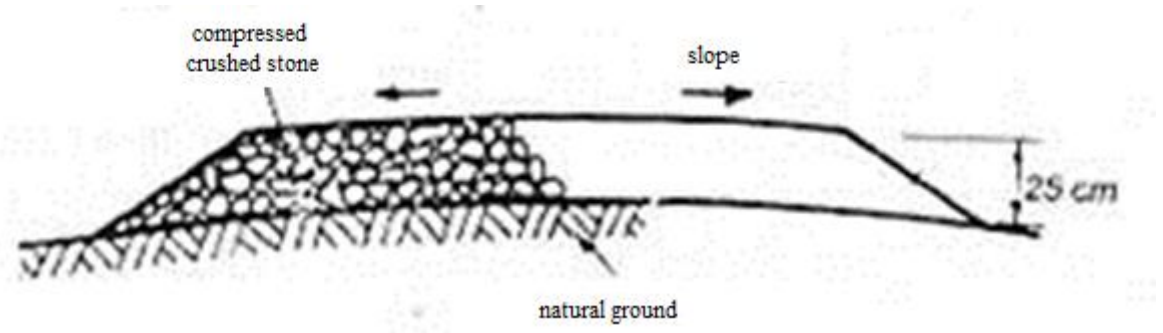
The first major development on road techique is the invention of the wheel around 5000 B.C. The first important roads on history are found at Mesopotamia around 3500 B.C. The first known road of history is constructed between Babylonia and Egypt around 3500 B.C. Mayan, Andean and Incan road remains have also been found.

The major militarian movements during Roman Empire resulted in developments on road construction technique. Roads constructed on that era are especially important because of their laminated structure. A typical Roman road is constructed having a width of 4,5 m. and having 4 layers. The thickness varies around 90-100 cm.

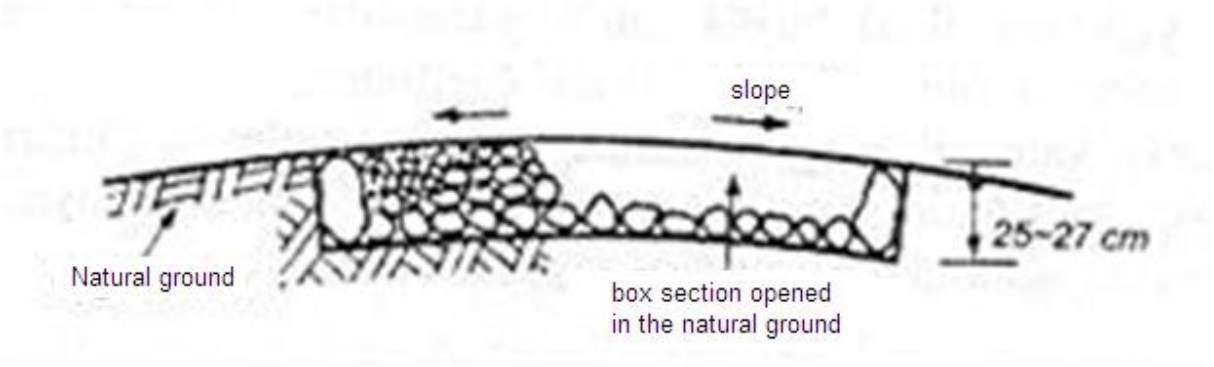


On faudal era, which is the result of Roman Empire's collapse, Roman roads have been neglected and ruined in a short time. During middle age, the condition of the European road network had been disastrous. The rise of European roads starts on the beginning of the 18th century.

Scotsman J. L. Mc-Adam (1756-1836) used the crushed rock on road construction. On Mc-Adam style roads, crushed rock, laid on levelled earth is compressed and a durable rolling surface is tried to be obtained. This kind of roads have been used until 1950's.



Another invention of the same period was made by French Pierre Tresaguet (1716-1796). Tresaguet dug the natural land to form a box section having the same width with the road. He placed bigger rocks on the base of this section and laid crushed rock on it. Another development of Mc-Adam and Tresaguet is the usage of the slope for surface drainage. Also the thickness of the road superstructure is decreased with respect to the Roman roads and descended to 20-25 cm.



On the other hand, Mc Adam saw that the ground has to be durable to face the load transferred from the pavement so that it can fulfill its task totally.

In the mid-19th century, major developments were seen on seaways and railways, including the use of steam machines. During this period, a halt has been seen in road construction. This situation continued until the automobiles became widespread at the beginning of the 20th century. Especially the entrance of automobiles into the transportation industry during the 1st World War recovered the industry. As Mc-Adam-type roads are suitable only for motorless vehicles, researchers have been focused on smoother rolling surfaces. On the other hand, as a result of the development of machine technology, bigger construction equipment has entered the industry, and a new era has begun. As machine power started to replace manpower, great economic benefits were gathered, and the current highway era began.

During the past 30 years between the end of the 20th and the beginning of the 21st century, some developments have been seen in the highway technologies. These developments, which originate especially from America, Europe, and Japan, have resulted in advanced highway networks. These roads, which have been constructed with very high-level quality standards, have also brought some problems as an inevitable result of the progression. Increasing vehicle traffic volumes have resulted in increases in traffic jams, delays, travel times, resource use,

environmental problems, and a number of accidents. To decrease or at least control this negative outcome of the transportation systems, systems are planned to be designed or operated in a more efficient, effective, safe, and economical way.

As a result of these studies, the "intelligent transportation systems" (ITS) concept has been put into use. Nowadays, intelligent transportation systems are the name of the technologies that are based on advanced technologies like computers, communication, and electricity; they use real-time or current databases that serve to solve operational, control, and management problems to increase effectiveness, efficiency, safety, and service quality on transportation. The concept of "intelligence" symbolizes the memory, communication, data analysis capability, and adaptability of the technologies. On the other hand, the integration of all transportation systems, which provides the carriage of men and goods from one place to another, is also included in the "intelligent transportation system" concept.

Some ITS Applications

- Highway management
- Incident management
- Arterial management
- Electronical fare collection
- Mass transit management
- Highway-railway intersection management
- Emergency management
- Regional multi-modal journey data

BRIEF HISTORY OF TURKISH HIGHWAYS

- **Oldest Roads**

Babylonia-Thapsacus road, which is an extension of the road network formed by Assyrians and Babylonians around 2000 B.C.

- **Hittites Roads**

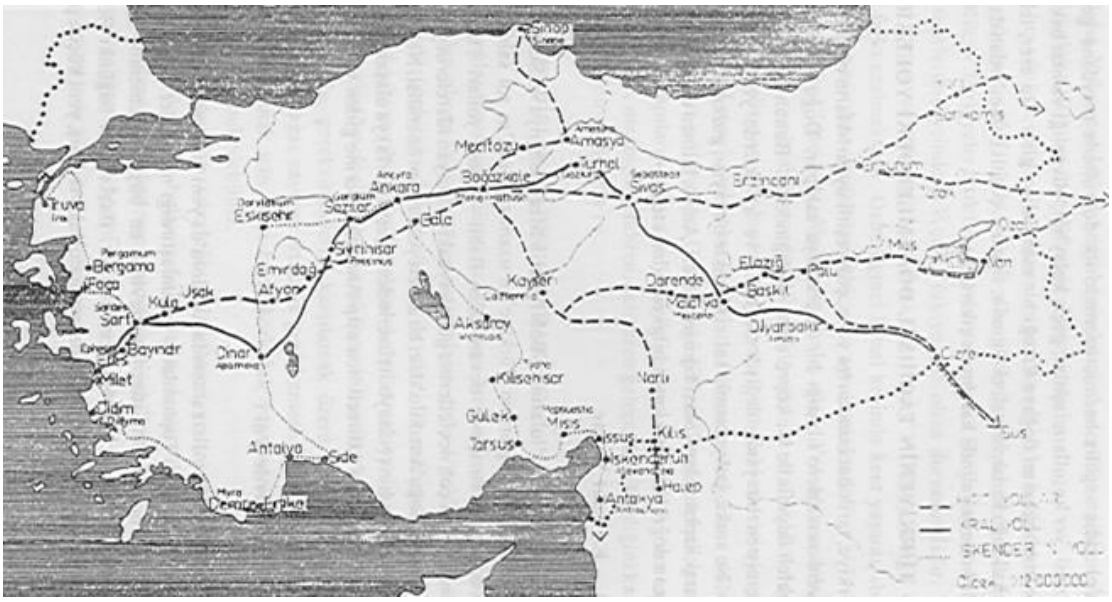
Hittites, Phyrigians, Lydians, Urartians which sovereigned Anatolia between 1700-2000 B.C., have built roads on Boğazkale, Sazılar, Sart, Van respectively and also built roads which direct to Iran, Mesopotamia, Syria, Black Sea and Aegean.

- **King's Road**

Persian king Darius who sovereigned Anatolia during the 6th century B.C. Constructed a road of 2.165 km between Sart (Aegean) and Sus (Northern Iraq). The best and the most secure road of this age have been named as the King's Road. King's Road, which passes Anatolia throughly have reduced the importance of the other roads. The importance of this road continued until Istanbul was made the capitol of the Eatern Roman Empire.

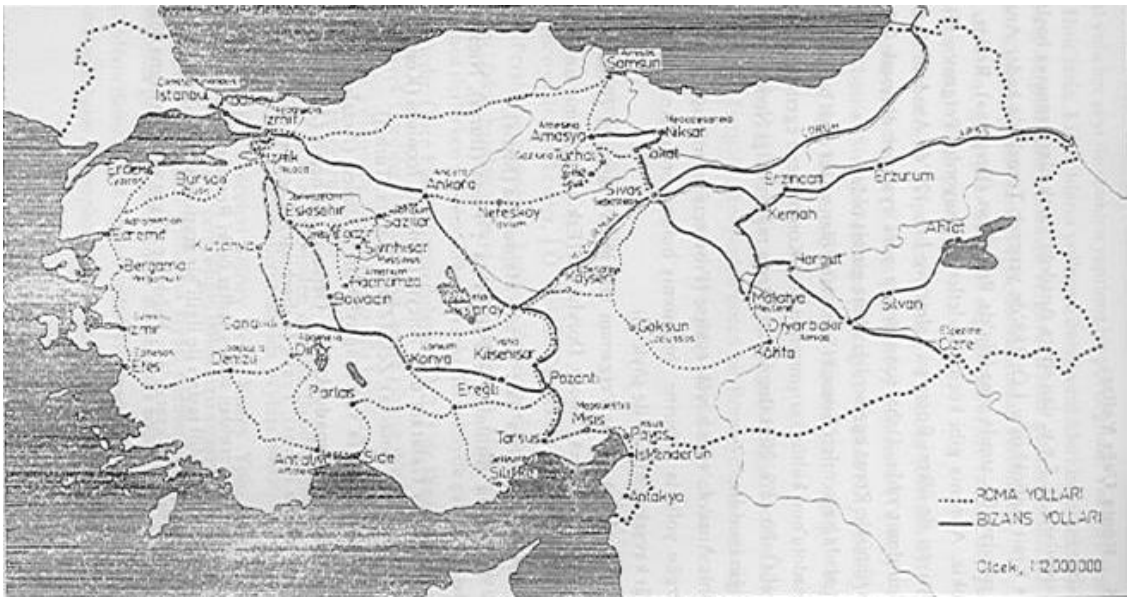
Hellenistic Age Roads

Alexander the Great, which entered Anatolia via Gallipoli Strait in 334 B.C., walked through South Aegean after reaching Pergamon and reached Ankara via Antalya, Side, Afyon, and Eskişehir. Then he reached Antioch via Aksaray, Tarsus, İskenderun. Alexander, who reached the Egypt has built a road network which mainly follows the coast.


Sinop
Merzifon
Amasya
Tokat
Turhal
Erzincan
Ankara
Bolu
Bolu
Eskişehir
Sivas
Gerede
Gölcük
Ilgaz
Kastamonu
Çankırı
Çorum
Yozgat
Kayseri
Darende
Elazığ
Bingöl
Muş
Van
Bergama
Foça
Kula
Uşak
Afyon
Emirdağ
Sarıkavak
Sarıkavak
Sarıkavak

Byzantine Age Roads

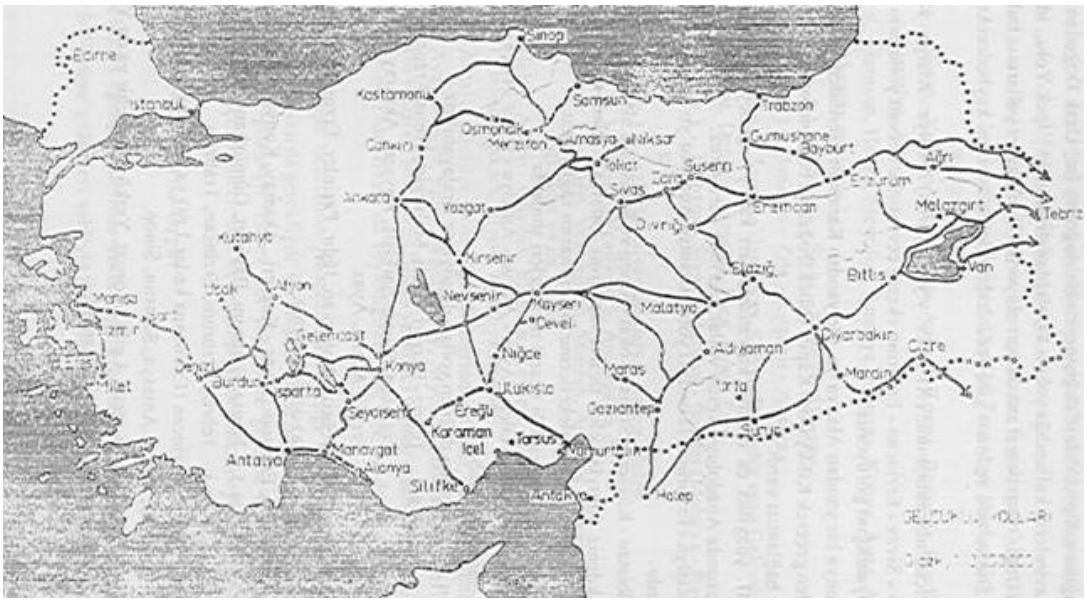
After the division of the Roman Empire, the Roman roads of Anatolia were still being used, but their starting points had changed. In the beginning, roads started from the Aegean region; during this age, first İzmit, then İstanbul became the start. Istanbul-İzmit-Konya road, constructed at this age, was then used by Ottomans and named the Pilgrimage Road.



Seljukian Age Roads

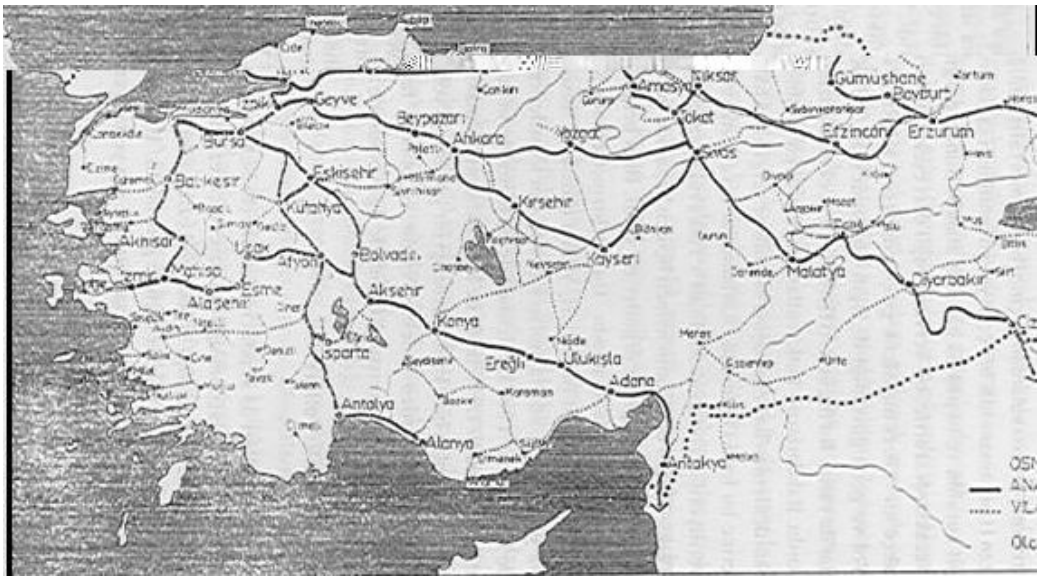
After Manzikert, the Turkish sovereign started in Anatolia (between the 11th and 14th centuries). During the Great Seljuk Empire, connecting Anatolia to the east was the main aim; however, during the Anatolian Seljuk Empire, Konya was the center of the road network. The caravan roads built around Antalya, Alanya, Konya, Aksaray, Sivas, Erzincan, and Erzurum axis gained importance. The only road that did not change its importance in this age was the Silk Road. Silk Road, which is an important arterial between west and east, preserved its importance until the exploration of the Cape of Good Hope.

Saljukians have paid great importance to roads and established the ribat organisation. Ribats were buildings constructed with the military intents but also served accomodation and religious services. Thay usually acted like foundations. Construction, management and protection of the roads were fulfilled by Ribat inhabitants. Ribats also formed the base of the caravanseraies.



Ottoman Age Roads

Ottoman roads are mainly the same as the Seljukian roads. However, they were improved with the changes worldwide. As a result of the improvement of seafaring in this age, roads in coastal cities have extended through the inner regions. Also, the exploration of the Cape of Good Hope and the opening of the Suez Canal changed the direction of the Ottoman roads. The roads that were neglected because of the increased importance of seafaring had totally been ruined after the improvements in railway technology during the 19th century.



During the rise of the Ottoman Empire, a road network that can answer the needs have been developed and this road network have been preserved for a long time. During the last period of the Empire, semi-military type road organisation have started to neglect the roads and the roads remained uncared. On this period some governors like Midhat Paşa and Halil Rıfat Paşa paid great importance on road construction.

The Republic of Turkey founded in 1923 has taken over 18.350 km. road network, 4.000 km. of which were in a good condition.

Ottoman roads have usually been constructed with military purposes or conquers. Also existing roads which remained from Seljukian, Byzantine or Roman ages have been used. For maintenance of these roads and to construct new roads for conquests, some financial organisations have been developed. Major organisations can be listed as: mountain pass (derbent) organisation, pavers (kaldırımcılar), seafarers (gemiciler), bridgers (köprücüler).

Republic Age Roads

In the first years of the Republic, railway construction, which was accepted as the most contemporary technology in that era, was important for the transportation sector. However, it was understood that just railway wasn't sufficient and highways were needed for the transportation system, so the Paved Ways and Bridges Presidency under the body of the Ministry of Public Works was founded in 1929 and highway construction works gained momentum with the road law in force.

After World War II, a new leap for the Highways was required due to the depression caused by the long war period and world economic crises.

The leap for highways was needed significantly in the year 1948. Until then, the target was to construct roads, disregarding the maintenance works. After 1948, it was underlined that road construction wasn't the only consideration; continuous maintenance of the existing roads was also a serious matter. It also indicated passing to a mechanized work period. Furthermore, the history of roads during the Republic period may be categorized as the period based on human power, the digging period before 1948, and the mechanized work period after 1948.

When the mechanized work period started on the construction of highways, the implementation of a fast and cheap system initiated gradually on the state highways and provincial roads. Besides, the inventory of current roads was prepared and a road network to meet the transportation need of Turkey was determined. Another important matter was to realize those works within the framework of specific plans and fixed programs.

With the requirement of directing these activities by a dynamic organization with contemporary methods, the General Directorate of Highways (KGM) was founded on March 1st, 1950. Thus, with the new highway policy, divisions under the body of the General Directorate of Highways were formed throughout the country. Necessary construction and maintenance equipment were provided to execute the mechanized work, and personnel training was given priority. Planning, survey and design, construction, and maintenance works have started.

General Directorate of Highways

General Directorate of Highways founded in 1950 aimed to reach the every corner of the country in the beginning. In that period roads with high standards weren't required due to the less traffic of motor vehicle.

Between the years 1950 and 1960, the first purpose of the work was to make the roads accessible in all weather conditions in order to take public services, such as health, education, etc., to every corner of the country. During this period, due to the gradual construction system, roads were improved according to the traffic density, new routes were specified, and regions of economic importance were connected with modern and appropriate road networks.

By implementing speed techniques, the road network expanded to 60 000 km. which met the need of Turkey in 1960. However, after it reached to a specific level, asphalt pavement became obligatory between the years of 1960 and 1970 due to the increase in traffic volume.

During the 1970s, with the establishment of the motor vehicle industry, the policy of highways gained new dimensions. Around the big cities and on the main axis where traffic jams occur, multilane expressways and motorway construction have became economical. A new technology was required due to the complexity of planning, financing, designing, and constructing this kind of road for which economic assessments were significant. General Directorate of Highways realized its first implementations with the project of the Bosphorus Bridge and İstanbul Ring Road.

Turkey, being aware of its geographic importance, spends great effort to establish transportation connections between Europe, Asia and Africa. The system of highway network in our country is significant to provide national development, unity and general development in the region.

The total length of the highway, which is under the control of the General Directorate of Highways, is 64 865 km. and is made up of three kinds of road:,

- Motorway
- State Highway
- Provincial Road

General Directorate of Highways is a public entity with special budget financing under the auspicious of the Ministry of Transport and Communication and is responsible for planning, design, construction, maintenance, and operation of motorways and state and provincial roads in compliance with the law numbered 6001. The roads under the control of other institutions are as follows:

- Village Roads
- Touristic Roads
- Forest Roads
- Urban Roads

The construction and maintenance of the touristic roads are carried out by General Directorate of Highways through the finance of the Ministry of Tourism. Village, Forest, Urban roads are respectively under control of Provincial Special Directorate of Administrations, The Ministry of Forestry and municipalities.

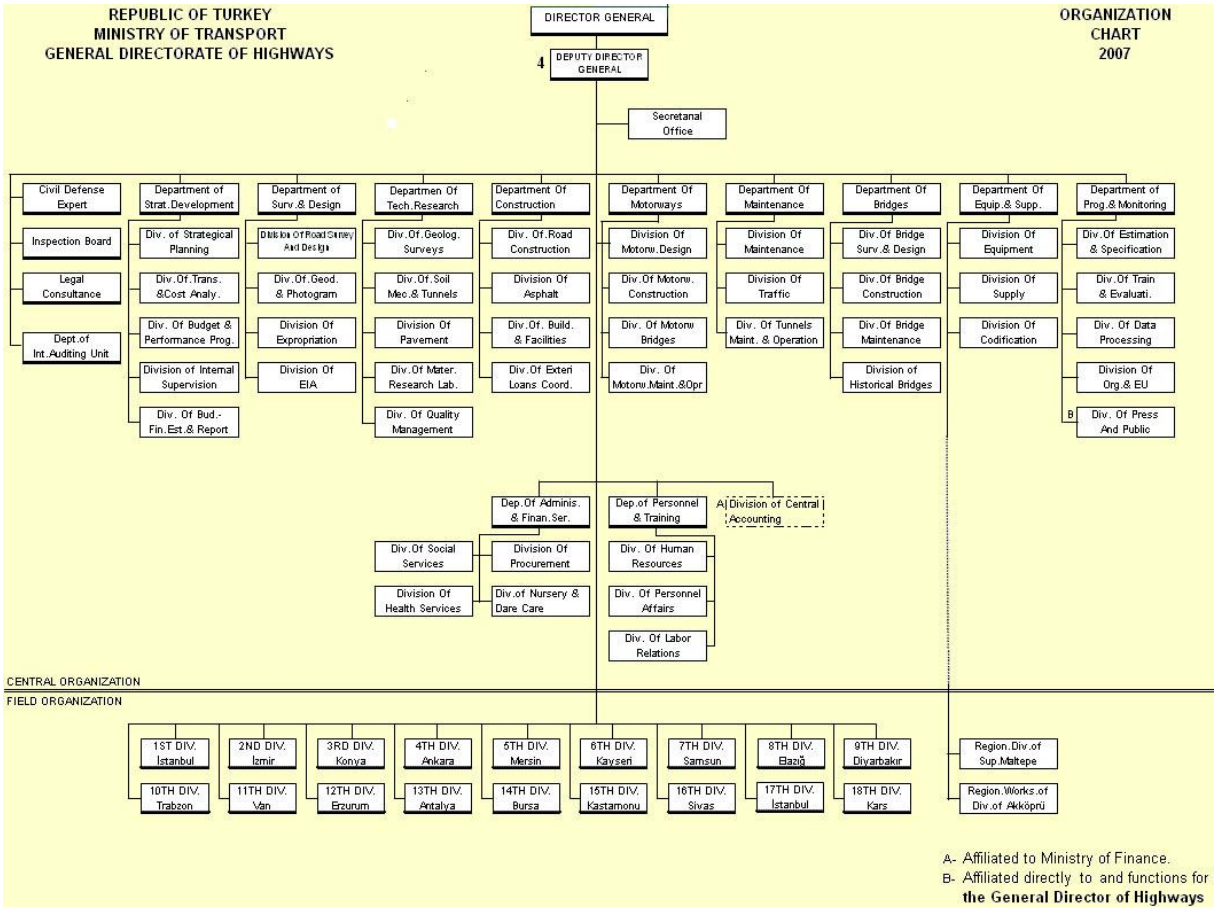
ROAD NETWORK ACCORDING TO SURFACE TYPES

ROAD NETWORK ACCORDING TO SURFACE TYPES (KM) 01.01.2011							
	Asphaltic Concrete	Surface Treatment	Stone Block	Stabilized	Earth	Primitive	Total
Motorways	2 080						2 080
State Highways	8 758	22 146	75	162	47	207	31 395
Provincial Roads	1 439	26 783	137	1 152	735	1 144	31 390
Total	12 277	48 929	212	1 314	788	1 351	64 865

Duties of the General Directorate of Highways

- To specify the network of motorways, state highways and provincial roads and to prepare the modifications on this network, to construct, improve, and repair the roads and bridges on the road network and keep them under continuous maintenance to provide the safe use of them and provide necessary training on these subjects,
- To design, to determine the standards concerning construction, repair, maintenance, and other issues, to prepare technical specifications,
- To determine the principles and rules about road and traffic safety, maintenance, and the use of roads, to realize them and to install the required road signs,
- To prepare the necessary maps for the works related to its duties, to carry out surveys and designs or to have them carried out,
- To collect, press, and publish the information about the activities of the General Directorate of Highways,
- To establish, maintain, repair, equip, operate, and rent garages and workshops necessary for the construction, improvement, maintenance, and safety of roads, warehouses, and storage for the equipment and materials, services and fuel stations, laboratories, test stations, bridges and all the other sub-foundations of historical road networks, required facilities for the landscape and the development of the roadside, roadside resorts, buildings to supply traffic safety and maintenance, wireless communication stations and required correspondence networks and all the other social facilities which will be useful for General Directorate of Highways to fulfill its duties sufficiently, or to have them established in accordance with the plans and designs prepared,
- In order to fulfill the defined duties, to determine, supply, store, and repair the required devices, materials, vehicles, equipment, and related parts and all the materials to operate and repair them, also to manufacture the necessary ones or to have them manufactured, to equip and operate the required warehouses, workshops and buildings,
- In compliance with the laws to expropriate, buy, rent, and occupy temporarily all the real estate and the buildings to provide safe and easy traffic flow, to fulfill the tasks included in the duties of the General Directorate of Highways,
- To perform the duties determined by the other laws related to motorways, state highways, and provincial roads.

General Directorate of Highways Organisation Chart



General Directorate of Highways Regions



BASIC CONCEPTS OF HIGHWAY ENGINEERING

Karayolu (*Highway,road*): Her çeşit kara taşıtı, yaya ve hayvanların yürürlükteki kurallara göre kullanmaları için oluşturulmuş ve kamunun yararlanmasına açık olarak yapılandırılmış olan arazi şeridir. (***A main public road, especially one connecting towns and citie; an open way (generally public) for travel or transportation.)***



Karayolu trafiği (*Highway traffic)*: Karayolunun ulaşım amacı ile her çeşit kara taşıtı (motorlu ve motorsuz) sürücüsü ile yayalar tarafından belirli kurallar çerçevesinde kullanılması sonucu üzerinde ortaya çıkan hareketliliktir. (***the movement of vehicles, people, etc, on highway for a particular purpose.)***

Otoyol (*Freeway, Motorway*): Sadece motorlu taşıtların kullanabileceği, kenar araziden sürekli bir korkulukla ayrılan, eşdüzey kavşak içermeyen, tam erişme kontrollü, iki platformlu ve yüksek standartlı karayoludur. (***A multilane, divided highway with a minimum of two lanes for the exclusive use of traffic in each direction and full control of access without traffic interruption.)***

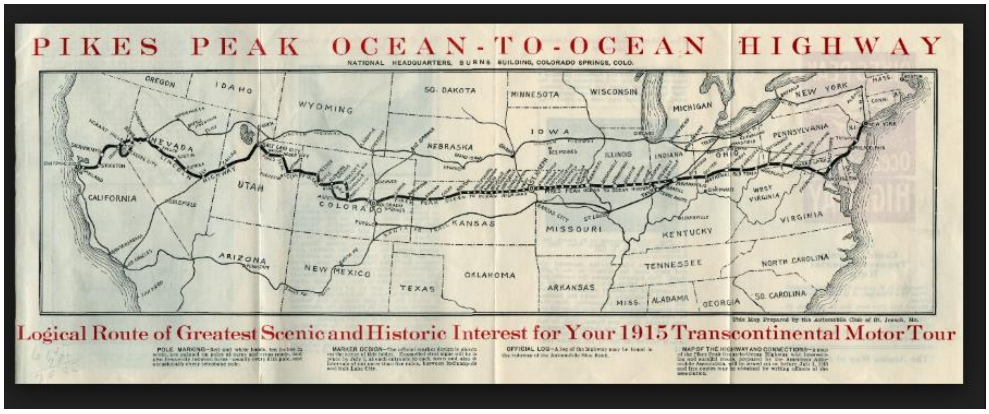


Proje hızı (*Design speed*): Bir karayolu projelendirilirken, geometrik tasarıma esas olan, sürücülerin emniyetli bir şekilde sağlayabilecekleri maksimum hız değeridir. ***(is a selected speed used to determine the various geometric design features of the roadway)***

Geçki, güzergâh (*Location, route*): Bir yolun arazi üzerinde (yeryüzünde) takip ettiği doğrultudur. Geçki düz ve eğri kısımlardan oluşur. (***The prescribed course to be traveled from a specific point of origin to a specific destination.)***



Geçki (Gülek Boğazı, Toroslar)



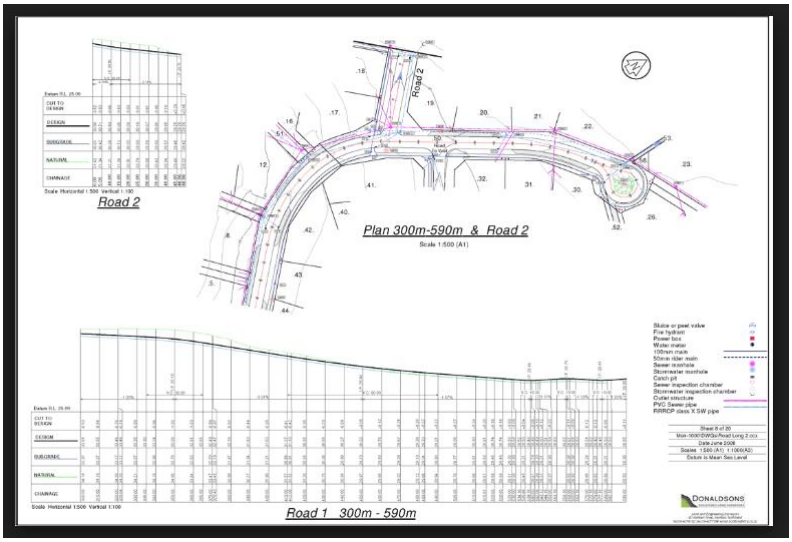


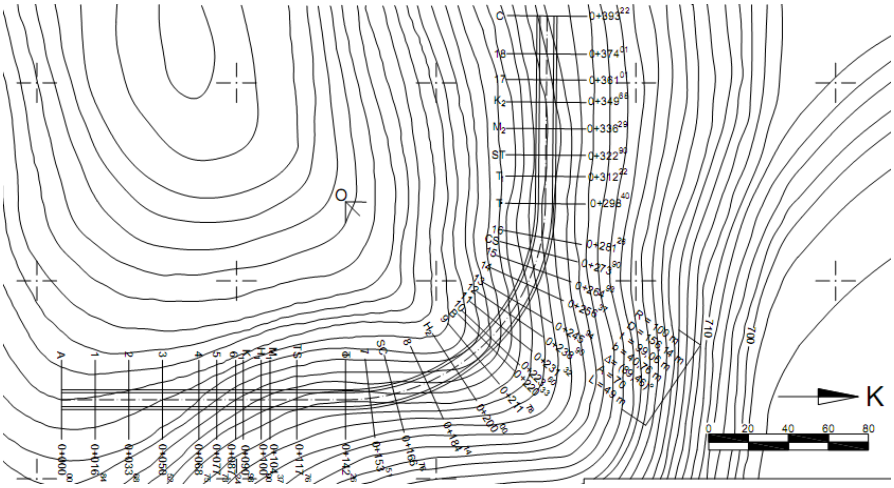
Geçki



Geçki

Plan *(Plan):* Tesviye eğrili bir haritada yolun bir yatay düzlem üzerindeki izdüşümüdür. Yani yolun kuşbakışı görünüşüdür.(***a drawing or diagram drawn on a plane; the projection of a road on a topographical map; scale drawing*** of a ***structure)***



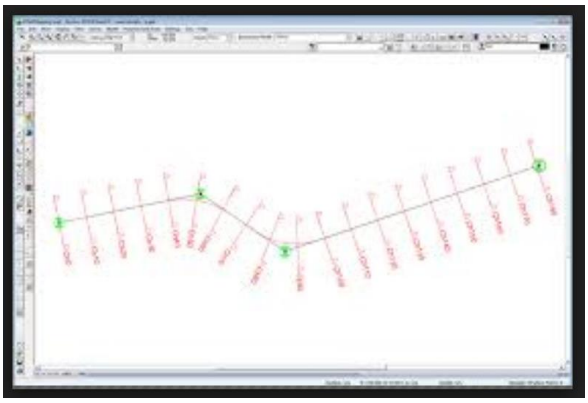


Plan

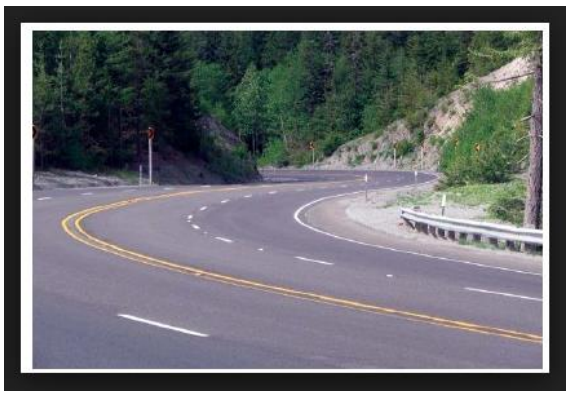
Yol ekseni *(Center line):* Planda yolun ortasından geçtiği kabul edilen hattır. (***a real or imaginary line that is equidistant from the surface or sides of something (road))***



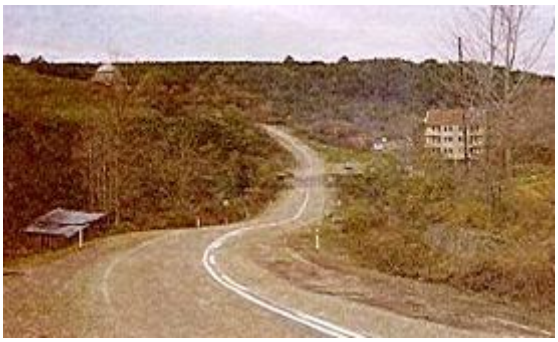
Aliyman (Alignment): Bir yol ekseninin planda düz giden kısımlarıdır. ***(The position or the layout of the central line of the highway)***



Yatay kurba (Horizontal curve): Bir yol ekseninin planda daire yaylarından oluşan eğri kısımlarıdır. ***(A horizontal curve provides a transition between two tangent strips of roadway, allowing a vehicle to negotiate a turn at a gradual rate rather than a sharp cut.)***



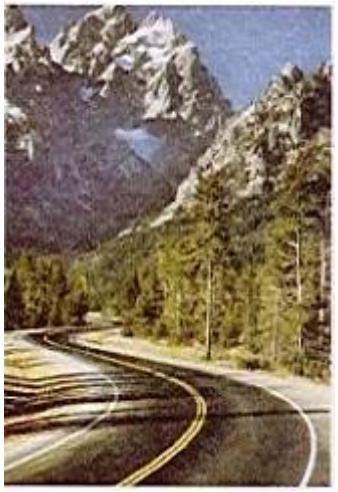
Düşey kurba (*Vertical curve*): Yol boykesiti üzerinde biribirini kesen farklı eğimdeki doğrular arasındaki eğri kısımlardır. Düşey kurbalar, dairesel ve parabolik olarak tasarlanırlar. (***curve that is applied to make a smooth and safe transition between two grades on a roadway or a highway.)***



A combination of a vertical and a horizontal curve



Combined horizontal curves



Reverse horizontal curves



Horizontal curves on double platform



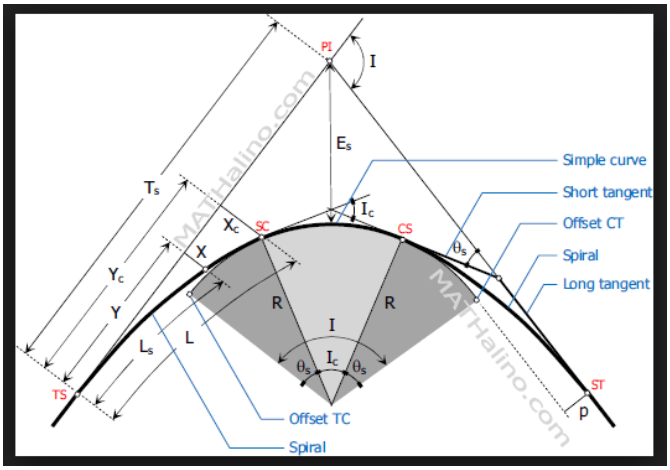
A horizontal curve on a viaduct



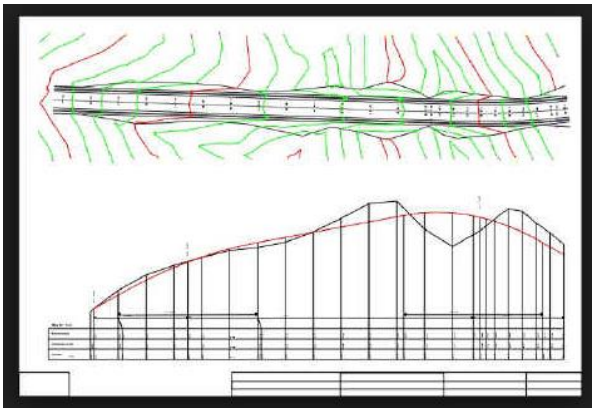
Vertical curve

Geçiş (Birleştirme) eğrisi, rakordman (*Transition curve*): Bir yol eksenindeki yatay kurbaların başlangıç ve bitişine konforu sağlamak maksadıyla yerleştirilen özel (Klotoid, lemniskat, kübik parabol gibi) eğrilerdir. ***(A curve, as on a highway, whose degree of curvature***

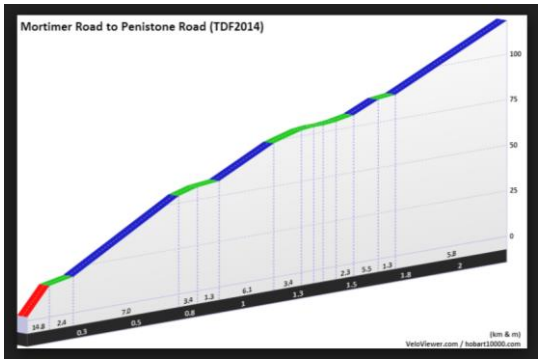
is varied to provide a gradual transition between a tangent and a simple curve, or between two simple curves which it connects. Also known as transition curve.)



Boykesit (*Profile*): Yol ekseninin boyuna doğrultudaki kesitinin açılımıdır. Bu açılımın düşey düzlemdeki izdüşümüyle arazinin tabii durumu ve arazi üzerindeki yolun son durumu görülür. (***A vertical section of the surface of the ground, or of underlying strata, or both, along any fixed line. On a highway, the profile is usually taken along the center line.)***

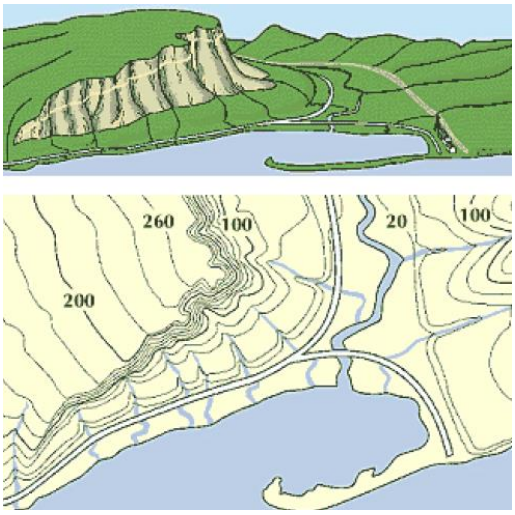


Kot, yükselti (*Elevation*): Dikkate alınan bir noktanın deniz yüzeyine veya bir kıyas düzlemine olan dik (şakulî) uzaklığıdır**. *(The elevation of a geographic location is its height above or below a fixed reference point)***



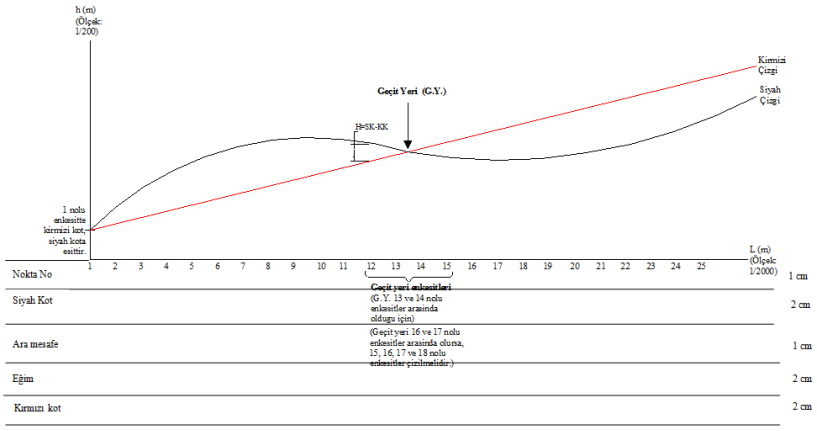
Tesviye eğrisi (*Contour line*): Düşey düzlemdeki kotları aynı olan yatay düzlemdeki noktaların üzerinden geçtiği kabul edilen sürekli eğrilerdir. Diğer bir deyişle, düşey düzlemdeki yerleri aynı olan yatay düzlemdeki noktaların, harita üzerindeki kesintisiz izdüşümleridir. (***a line on a map or chart joining points of equal height or depth.)***

Tesviye eğrili harita (*Contour map*): Bir bölgenin tesviye eğrileriyle tanımlandığı haritasıdır. Karayolu projelerinin yatay tasarımı tesviye eğrili haritaların üzerinde yapılır.*(a* ***topographic map*** *on* ***which the shape*** *of* ***the land surface*** *is* ***shown*** *by* ***contour lines, the relative spacing*** *of* ***the lines indicating the relative slope*** *of* ***the surface)***



Siyah çizgi (*Ground line*): Bir yol boykesitindeki tabii zeminin durumunu kırıklı çizgi şeklinde gösteren hattır. ***(the line representing ground level)***

Siyah kot (*Ground elevation*): Yol boykesitindeki bir noktaya ait tabii zeminin kotudur. Diğer bir deyişle siyah çizgi üzerindeki herhangi bir noktanın kotudur. ***(The elevation of a point on the ground level; in other words, the elevation of a point on the ground line)***



Kırmızı çizgi (*Grade line*): Toprak işi sonunda yol ekseninin boykesitteki halini gösteren doğru ve eğri kısımlardan meydana gelen hattır. (***a longitudinal reference line or slope to which a highway or railway is built)***

Kırmızı kot (*Grade elevation*): Boykesitteki bir noktaya ait yolun bitmiş durumundaki kottur. Diğer bir deyişle kırmızı çizgi üzerindeki herhangi bir noktanın kotudur. ***(the elevation of a point on the profile of a built road; in other words, the elevation of a point on the grade line)***

Toprak işi, tesviye (*Earth work, grading*): Bir karayolu projesinin uygulanması sırasında tabii arazi yüzeyinin belirlenen şekillerde değiştirilip, yol gövdesi ortaya çıkarılırken zemin üzerinde yapılan kazma (yarma), doldurma (dolgu), taşıma (taşın) ve sıkıştırma ile ilgili işlemlerin bütünüdür. ***(an embankment or other construction made of earth; especially, one used as a field fortification; the operations connected with excavations and embankments of earth;)***



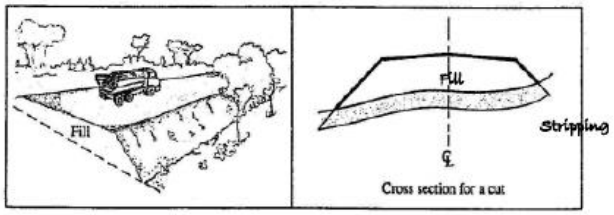
Yarma (*Cut*): Toprak işi sırasında yolun kırmızı çizgisi üzerinde kalan ve kazılması gereken kısımdır. Yarmadan çıkan toprak ya dolguda kullanılır ya da depoya gönderilir. ***(The part of earth above the grade line of a road that has to be cut. The cut part is used as fill or sent to storage)***

Dolgu (*Fill*): Toprak işi sırasında yolun kırmızı çizgisi altında kalan ve doldurulması gereken kısımdır. Dolguya ilave edilen malzeme ya yarmadan temin edilir ya da bir ocaktan ödünç olarak alınır. ***(The part of earth under the grade line of a road that has to be filled. The fill part is provided from the cut part road or from the quarry)***

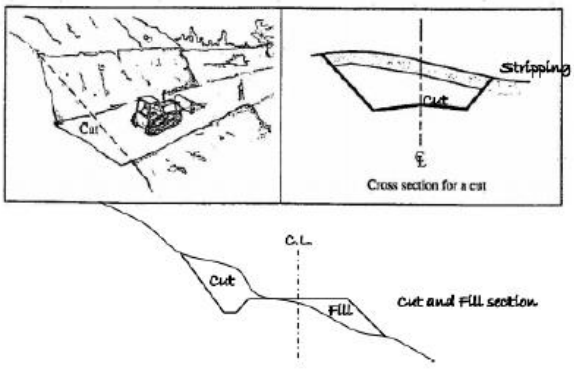








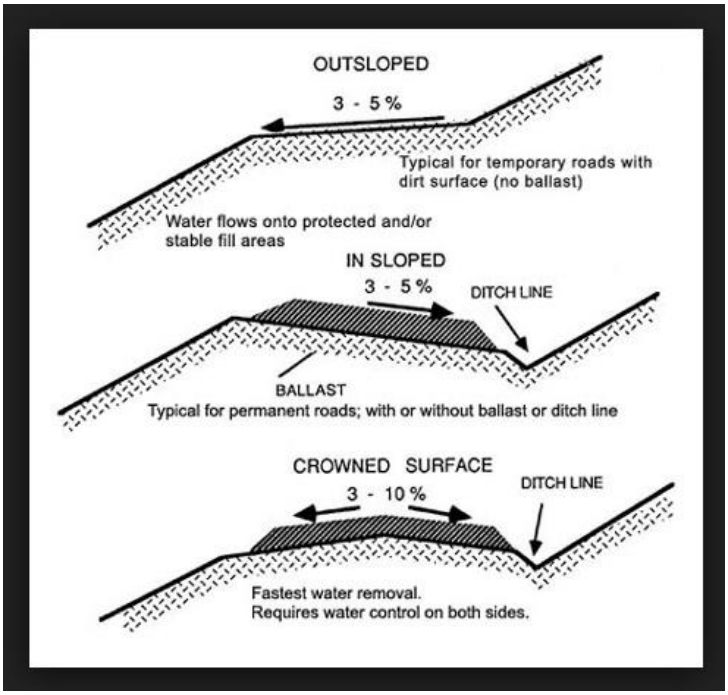
The (+ve) sign indicates an excavation.



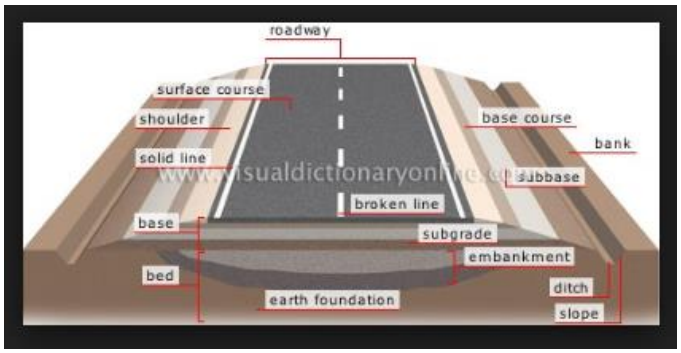
Tesviye yüzeyi (*Formation*): Toprak işi sonunda ortaya çıkan ve üstyapı için hazır hale getirilmiş taban yüzeyidir. ***(The surface of the base built by the earthwork which is prepared for the superstructure.)***

İnce tesviye, reglaj (*Finishing*): Toprak işi bitirilirken, yolun istenilen enine ve boyuna eğimi alması için en son olarak yapılan tesviye işlemidir. Reglajdan sonra hemen üstyapının inşaasına geçilebilir.

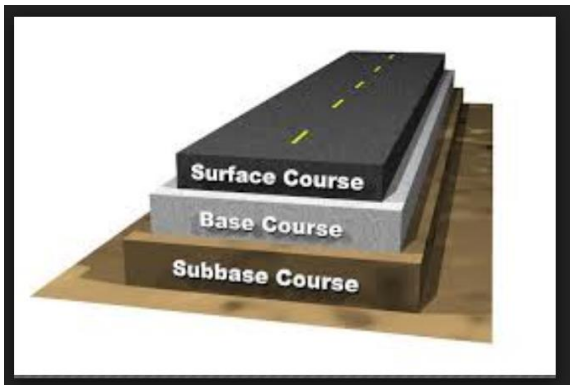
Drenaj (*Drainage*): Yeraltı suları ile yüzeysel suların, yol tesislerine zarar vermeden, en uygun bir şekilde uzaklaştırılması için yapılan işlemlerdir. ***(To draw off the ground water without harming the road facilities; the act of draining)***



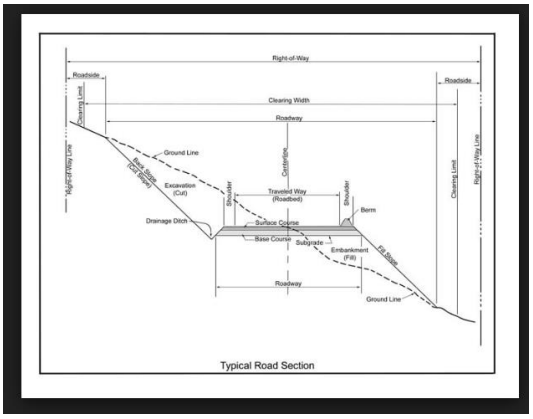
Altyapı (*Sub-grade*): Yolun toprak işi sonunda, projesinde belirlenmiş kot ve enkesit şekline getirilen kısmıdır. Üstyapının maruz kaldığı bir kısım etkiyi altyapı karşılar. Drenaj yapıları, köprü, menfez, tünel, viyadük, istinat ve iksa duvarları, tahkimat gibi sanat yapıları, şevler altyapı içinde değerlendirilir. (***The level layer of rock or earth upon which the foundation of a road or railway is laid; the ground beneath a roadway or pavement)***



Üstyapı (*Pavement*): Karayolu üzerindeki trafik yüklerinin doğurduğu etkilerin bir bölümünü karşılamak, bir bölümünü ise emniyetli bir şekilde altyapıya aktarmak için altyapının üzerine genellikle alttemel, temel ve kaplama tabakası olarak inşa edilen kısımdır. ***(The combination of subbase, base, base course and surface course placed on a subgrade to support the traffic load and distribute it to the road bed.)***



Enkesit (*Cross-section*): Yol ekseninin herhangi bir noktasından, eksene dik olarak belirli bir doğru boyunca alınan kesittir. (***a vertical section of the road axis)***





Fill cross-section

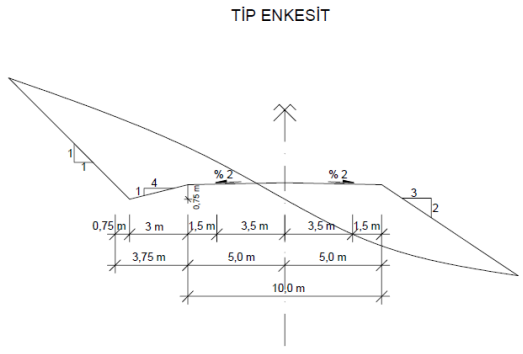


Cut cross-section



Combined cross-section

Tip enkesit (*Typical cross-section*): İnşa edilecek bir yolun, karar verilmiş standart ölçülerini ve özelliklerini ayrıntılı olarak gösteren enkesittir. Tip enkesitte, plâtform, kaplama, şerit, banket, yarma hendeği, varsa röfüj boyutları; enine eğim, yarma ve dolgu şevlerinin eğimleri; üstyapı tabakaları, kalınlıkları ve malzeme cinsleri; kamulaştırma genişliği belirtilir.



Trafik şeridi (*Traffic lane*): Taşıtların karayolu boyunca bir dizi halinde güvenle seyretmeleri yanında küçük enine yer değiştirme manevralarına da imkan veren yeterli miktardaki yol genişliğidir. (***any of the parallel strips into which the carriageway of a major road or motorway is divided)***



Banket (*Shoulder*): Yol kaplamasının dış tarafı ile şev başlangıcı arasındaki kısımdır. Yolcu ve yük indirip bindirmek ve çeşitli nedenlerle duraklamak için kullanılmasının yanında kaplamayı yandan destekleyip bozulmasını geciktirmek görevleri de vardır. Yerine göre granüler veya bitümlü malzemeden yapılabilir. ***(A portion of the roadway contiguous with the traveled way for accommodation of stopped vehicles, emergency use, and lateral support of the subbase, base, and surface courses.)***



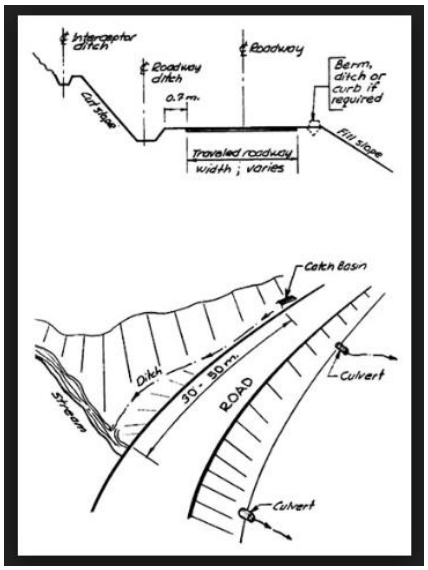
Platform (*Roadway*): Yolun enine yönde kaplama ve banketlerden oluşan bölüntüsüz kısmıdır. Bölünmemiş yollar tek platformlu, bölünmüş yollar çift platformlu olarak anılır. (***a road (especially that part of a road) over which vehicles travel; carriageway; one of the two sides of a motorway where traffic travels in one direction only usually in two or three lanes)***

Kaplama tabakası (*Surfacing, pavement*): Yol platformunun, banketler arasında kalan ve temel tabakası üzerine inşa edilmiş, bitümlü karışımlar, beton veya parke taşlarından yapılmış, taşıtlara uygun bir yuvarlanma yüzeyi teşkil etmek üzere ayrılmış kısmıdır. (***the hard layered structure that forms a road carriageway, airfield runway, vehicle park, or other paved areas)***

Temel tabakası (*Base course*): Alttemel ve kaplama tabakası arasına yerleştirilen, belirli granülometrik ve dayanım özellikleri olan kum, doğal çakıl ve kırma taş ile bir miktar bağlayıcı ince malzemeden yapılan, trafik yükünün etkilerini taban zeminine yayarak aktarmak için serilen tabakadır. (***A layer of selected material of planned thickness, constructed on the subgrade or subbase for the purpose of serving one or more functions such as distributing load, providing drainage, minimizing frost action, etc.)***

Alttemel tabakası (*Sub-base course*): Taban zemini ile temel tabakası arasına (Altyapının hemen üzerine) yerleştirilen, temel tabakasına göre daha düşük kalitedeki kum, çakıl, taş kırığı, yüksek fırın cürufu gibi daneli malzemenin kullanılmasıyla oluşturulan, trafik yüklerinin taban zeminine aktarılmasında temel tabakasına yardımcı olan, su ve don etkilerine karşı üstyapının korunmasına katkı sağlayan tabakadır. (***For the support of paving materials, a layer of granular material applied to the subgrade in a predetermined depth; provides underpavement drainage and reduces frost damage.)***

Kafa hendeği (*Intercepting ditch*): Yolun yarma kesimlerinde, yağış suları şevin bozulmasına ve erozyona sebep olacak düzeyde ise, yarma şevinin üst kısmına yapılan hendektir. Buralarda toplanan su ya kenar hendeğine akıtılır ya da yakınlardaki tabii bir su yatağına yönlendirilir**.** ***(An interceptor ditch is a small ditch or channel constructed to intercept and convey water to an area where it can be safely discharged.)***



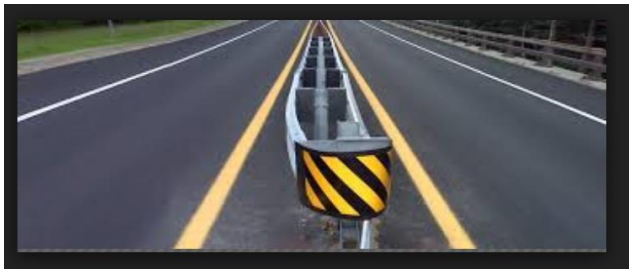
Kenar hendeği (*Ditch*): Yol platformunun kenarı ile yarma şevi arasında bulunan ve buralardan gelen yüzeysel suların toplanıp uzaklaştırılması için yapılan kanallardır. Genellikle üçgen ve yamuk kesitli olarak inşa edilir. (***A small artificial channel cut through earth or rock to carry water for irrigation or drainage. A long narrow cut made in the earth to bury pipeline, cable, or similar installations.)***

Kamulaştırma genişliği (*Right-of-way)*: Bir geçki boyunca yol ile ilgili bilumum tesislerin bulunması için oluşturulmuş arazi şeridi dir. Bu genişliğe dahil olan arazilerdeki tüm mülkler sahiplerinden satın alınarak kamulaştırılır ve kamulaştırma sınırları kurallar çerçevesinde herkesin kullanımına açıktır. ***(the legal right of someone to pass over another's land, acquired by grant or by long usage; the path or road used by this right)***

Ayırıcı, röfüj (*Refuge*): Yoldaki trafik akımlarını ayırmak ve yayalara güvenli bir alan sağlamak maksadıyla plâtform içinde oluşturulan farklı kottaki kısımdır.



Orta ayırıcı, orta röfüj (*Median, central reserve*): Bölünmüş yollarda platformları birbirinden ayıran ve eksen hattı boyunca kaplamadan farklı kotta inşa edilen kısımdır. Yerine göre ağaçlandırılabilir veya drenaj için de kullanılabilir. (***A median is the portion of a roadway separating opposing directions of the traveled way.)***



Kenarlık, bordür (*curb, kerb*): Kentiçi yollarda kaplama ile yaya kaldırımı arasına, bölünmüş yollarda kaplama ile orta röfüj arasına yerleştirilen, doğal taş, beton gibi rijit malzemelerden yapılan ve kaplamayı sınırlandıran yol elemanıdır. ***(Curbs, berms, and edging are roadside elements, usually constructed of granite or extruded bituminous concrete used to define the pavement edge and to control drainage.)***



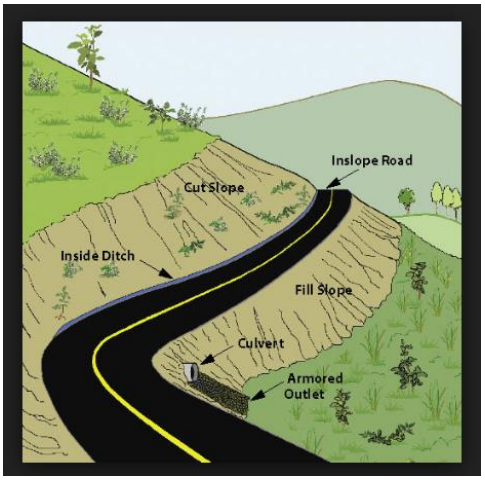
Bordür oluğu, kanivo (*gutter*): Yollarda kaplama ve yay kaldırımındaki yüzeysel suların toplanıp uzaklaştırılmasına yarayan küçük kanaldır. ***(A channel at the edge of a street or road for carrying off surface water.)***



Baca, Rögar (*Manhole*): Bordür oluğunda toplanan yüzeysel suların atıksu veya yağmur suyu kanalına aktarılması için belirli aralıklarla inşa edilen drenaj yapısıdır. ***(a shaft with a removable cover that leads down to a sewer or drain; a hole, usually with a detachable cover, through which a man can enter a boiler, tank, etc)***



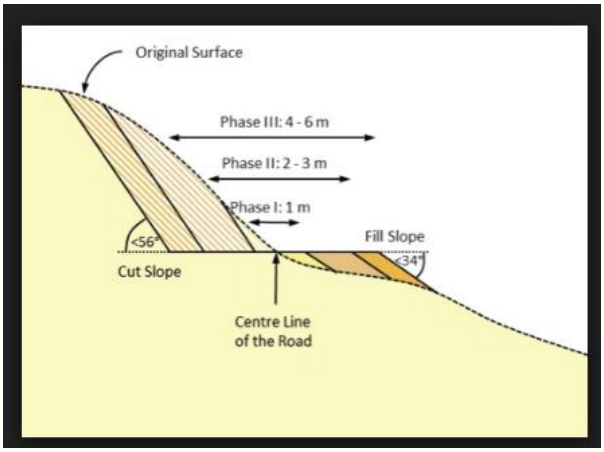
Şev (*Side slope*): Bir yol enkesitindeki, yarma kesimlerinde kenar hendeği tabanı ile tabii zemin arasında; dolgu kesimlerinde platformun dış kenarı ile tabii zemin arasındaki eğik yüzeydir. İhtiyaca göre çeşitli eğimlerde yapılır.



Şev kazığı (*Slope stake*): Dolgu ve yarma şevlerinin tabii zemini kestiği etek ve kret noktalarının yerini belirtmek için kullanılan kazıktır. (***Slope staking is a trial and error process of finding the point where the slope intercepts the natural ground. This point is referred to as the "catch point" or "slope catch.")***

Enine eğim (*Slope, crossfall*): Yol platformuna gelen yüzeysel suların bir an önce uzaklaştırılması maksadıyla, yol eksenine dik doğrultuda her iki yana verilen eğimdir. Genellikle banket eğimi drenaj amacıyla kaplama eğiminden daha fazla yapılır. (***Surface cross slopes are necessary on all components of the crosssection to facilitate drainage.)***

Eksen çizgisi (*Center line*): Bölünmemiş yol platformunda karşılıklı trafik akımlarını kullanacakları genişliği göstermek için genellikle yolun ortasına boyuna doğrultuda çizilen çizgidir***. (A painted line running along the center of a road or highway that divides it into two sections for traffic moving in opposite directions or, in the case of a divided highway, for lines of traffic moving in the same direction at different speeds.)***



Boyuna eğim (*Grade*): Yol ekseni doğrultusunda (yola boyuna doğrultuda) verilen eğimdir. Yüksek standartlı yollarda düşük boyuna eğim uygulanır.(***The degree of inclination of a slope, road, or other surface; A slope or gradual inclination, especially of a road or railroad track:)***

Şerit çizgisi (*Lane line, traffic line*): Aynı yöndeki trafiğin kullanacağı şeritleri birbirinden ayırmak maksadıyla yolun boyuna doğrultusunda eksen çizgisi haricinde çizilen çizgilerdir.

Kenar çizgisi (*Edge line*): Yolun kaplama kenarlarını göstermek için kaplamanın her iki yanına boyuna doğrultuda çizilen sürekli çizgilerdir.

Park şeridi (*Parking lane*): Yüksek standartlı yollarda öncelikle taşıtların park etmeleri için ayrılmış bir yardımcı şerittir.



Tırmanma şeridi (*Climbing lane*): Dik eğimli bir yol kesiminde kapasite ve hizmet düzeyini artırmak maksadıyla öncelikle yavaş giden taşıtların kullanmaları için, eğim yukarı yönde, yolun en sağına inşa edilen yardımcı şeritlerdir. ***(A passing lane added on an upgrade to allow traffic to pass heavy vehicles whose speeds are reduced.)***



Bölünmüş yol (*Divided road, dual carriageway*): Bir yöndeki trafiğin karşıdan gelen trafikle orta röfüj, korkuluk ya da başka bir fizîkî engelle ayrıldığı, iki platformlu yoldur. Genelde şehiriçinde veya yüksek standartı sağlamak için uygulanır. ***(a highway divided down the middle by a barrier that separates traffic going in different directions)***



Bölünmemiş yol (*Undivided road*): Üzerinde karşı yönden gelen trafiği ayıran herhangi bir fizîkî engel bulunmayan, tek platformlu yoldur.



Korkuluk, bariyer (*guard fence, safety fence*):Taşıtların tehlikeli durumlarda yoldan çıkmalarını veya orta röfüjü geçip karşı yöne girmelerini önlemek için platform kenarlarına konulan beton, çelik profil veya halattan imal edilmiş engellerdir.

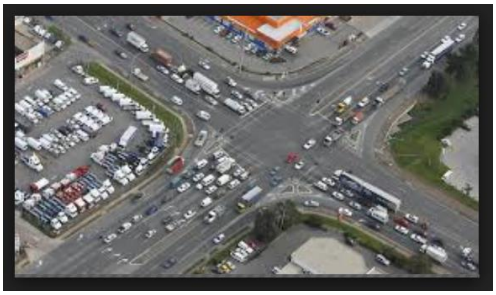


Kenar taşı (*Marker post, delienator*): Yolun doğrultusunu ve sınırlarını belirlemek için, platform kenarlarına belirli aralıklarla yerleştirilen, gece görüşünde kolaylık sağlaması için üzerinde ışık yansıtıcılar da bulunan güvenlik elemanlarıdır.



Eşdüzey kavşak (*Intersection, junction*): Yolların kesiştiği bir alanda, trafik akımlarının aynı mekanı ayrı zamanlarda kullandıkları durumdaki kavşaktır. Ortak kullanılan alanda, farklı doğrultudaki yolların yüzeyleri aynıdır. (***the place or point where two or more things come together;*** **especially** ***: the place where two or more streets meet or cross each other)***

Kavşak (*Intersection*): Farklı yön ve doğrultudan gelen trafik akımlarının önceden belirlenmiş kurallara uygun bir sırayla (düzenle) kullandıkları yol parçalarıdır.



Katlı kavşak (*Interchange*): Yolların kesiştiği bir alanda, trafik akımlarının ayrı mekanları aynı anda kullanmasıyla ortaya çıkan kavşak çeşididir. Ortak kullanılan alanda her yol farklı yüksekliktedir ve kendi aralarındaki bağlantılar yan yollar vasıtasıyla yapılır. ***(A highway intersection allowing traffic to move freely from one road to another without crossing another lane of traffic.)***



Dönel kavşak ***(Roundabout)***: Ortasında trafiğin etrafında döndüğü yuvarlak bir ada bulunan kavşaklardır. ***(An unsignalized intersection with a circulatory roadway around a central island with all entering vehicles yielding to the circulating traffic.)***

